Today's agenda

- ↳ Chars and String
- ↳ ASCII
- ↳ Problems

int[]: Collection of integers

String: Collection of Characters

ex: "AlgoPrep"

Characters:

a) Alphabet

↳ a-z (lowercase)

↳ A-Z (uppercase)

b) Special characters

↳ @ # $? ! etc.

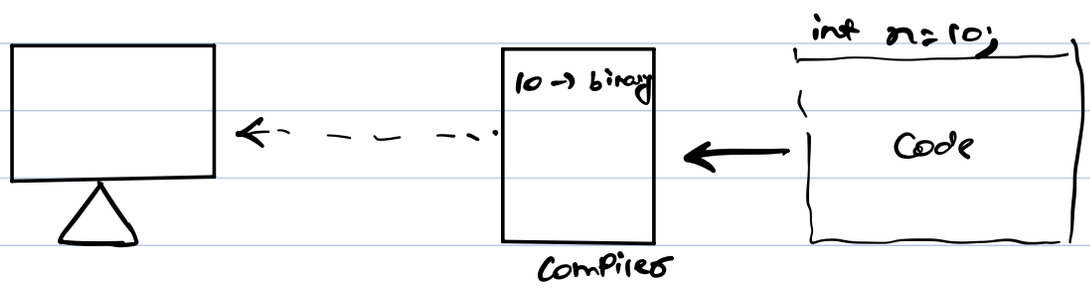
c) Numbers

↳ 0, 1, 2 9

Syntax:

```
char  ch = 'A';
```

(char: type, ch: name, 'A': character)



Characters → numbers → binary numbers

(Characters → numbers: ASCII values; predefined)

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Char ch = 'B'; ⇒ 'B' = 66

ASCII → 256 Characters

'A' : 65	'a' : 97	'0' : 48
'B' : 66	'b' : 98	'1' : 49
'C' : 67	'c' : 99	⋮
⋮	⋮	'9' : 57
'Z' : 90	'z' : 122	'10' : Not a Character

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* char Rules

1. you can do any mathematical operation on character & the answer will be integer.

s.o.p ('A' + 'B'); → 131

(65, 66)

typecast

2. char to int : implicit

ex: int x = 'c';

3. int to char : Complicated → in few cases it will be implicit
↳ in few cases it should be explicit

ex: char ch = 65;

do explicitly always.

```
// "static void main" must be defined in a public class.
public class Main {
    public static void main(String[] args) {
        Scanner scn = new Scanner(System.in);
        char ch1 = 'A';

        //int x = scn.nextInt();
        char ch2 = scn.nextLine().charAt(0);
        System.out.println(ch2);


    }
}
```

Quiz 1:

```
Char ch1 = 'B';
S.O.P (ch1);   - B
```

Quiz 2:

```
int x = 'A';
x = x+2;
S.O.P (x);   → 67
```

67

~~65~~

x

Quiz 3:

```
Char ch3 = 'XYZ';   ⇝ error
S.O.P (ch3);
```

Quiz 4:

```
Char ch2 = (char) 66;
S.O.P (ch2);   → 'B'
```

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Quiz 5:

```
char ch4 = 'A';
ch4 = (char) ch4 + 3;
S.O.p(ch4);
```

65 → 68

→ error

Quiz 6:

```
char ch5 = 'A';

if (ch5 >= 90){
    S.O.p("greater");
}
else{
    S.O.p("smaller");
}
```

→ Smaller

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// Strings

↳ Collection of characters.

Syntax:

↳ String St = "AlgoPrep";

String St1 = "A"; → this works

St

	0	1	2	3	4	5	6	7
→	A	L	G	O	P	R	E	P

→ S.o.p (St.charAt(3)); → O

→ St.~~charAt~~(3) = ~~'Y'~~'z'; → error

↳ In String you can't change characters directly. (no update).

↳ Strings are immutable.

→ **SubString** → Any Continuous Part of String

String St = "AlgoPrep";

↳ Algo ✓ ↳ goPrep ✗

↳ A ✓

↳ Prep ✓

↳ AlgP ✗

↳ AlgoPrep ✓

```java
    public static void main(String[] args) {
        Scanner scn = new Scanner(System.in);
//          char ch1 = 'A';

//          //int x = scn.nextInt();
//          char ch2 = scn.nextLine().charAt(0);
//          System.out.println(ch2);

        String st1 = "AlgoPrep"; //lgopre
        System.out.println(st1.charAt(3)); // -> o

        // String st2 = scn.nextLine();
        // System.out.println(st2);

        //substring -> continuous part of string

        System.out.println(st1.substring(1,7)); // sp - inclusive, ep - exclusive

        //length of string
        System.out.println(st1.length());

        // String to char Array
        char[] arr = st1.toCharArray(); //AlgoPrep
        for(int i=0;i<arr.length;i++){
            System.out.print(arr[i]+" ");
        }

        //Char Array to String
        String dhsbhaiuhja = "".valueOf(arr);
        System.out.println(dhsbhaiuhja);
```

Break till 9:48 PM

Q) Toggle Characters

↳ Given a char[] which contains only small and capital letters, toggle them.

↳ lowercase → uppercase

↳ uppercase → lowercase

ex: AlgoPrEP → aLGOpRep

'A' :	65	$\xrightarrow{+32}$ / $\xleftarrow{-32}$	'a' :	97
'B' :	66	$\xrightarrow{+32}$ / $\xleftarrow{-32}$	'b' :	98
'C' :	67	$\xrightarrow{+32}$ / $\xleftarrow{-32}$	'c' :	99
'2' :	90			

uppercase to lowercase : +32

lowercase to uppercase : -32

→ wherever you want to change char, you think in terms of ascii.

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//Psuedo code

```
P s void toggle (char[] ch){

    for (int i=0; i<ch.length; i++) {
        if (ch[i] >= 65 && ch[i] <= 90){

            ch[i] = (char)(ch[i] + 32);
        }
        else{ //lower case

            ch[i] = (char)(ch[i] - 32);
        }
    }
}
```

T.C: O(n)

S.C: O(1)

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Q) Reverse the given String

↳ Given a String str, reverse the String.

//Pseudo code

```
Public static String reverseString (String str){

    Char[] ch = str.toCharArray();

    int sP = 0;
    int eP = ch.length -1;

    while (sP< eP) {
        char temp = ch[sP];
        ch[sP] = ch[eP];
        ch[eP] = temp;
        sP++;
        eP--;
    }

    return "".valueOf(ch);
}
```

T.C: O(N)

S.C: O(N)

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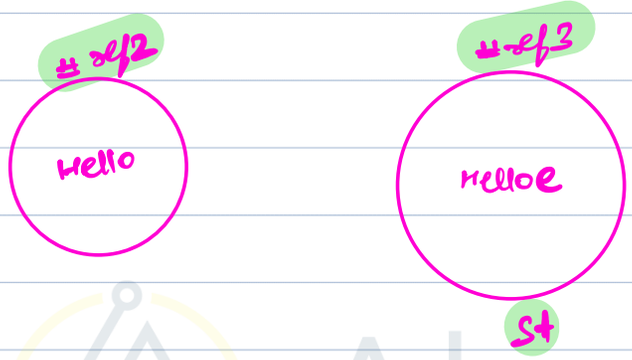
→ String St = "Hello"; → immutable

St = St + "e"; → T.C: O(N)

S.O.P (St); → Helloe

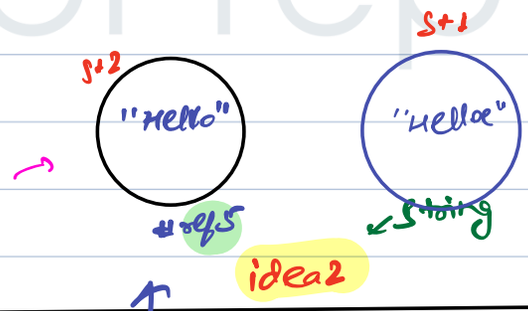
↳ concatenation can happen

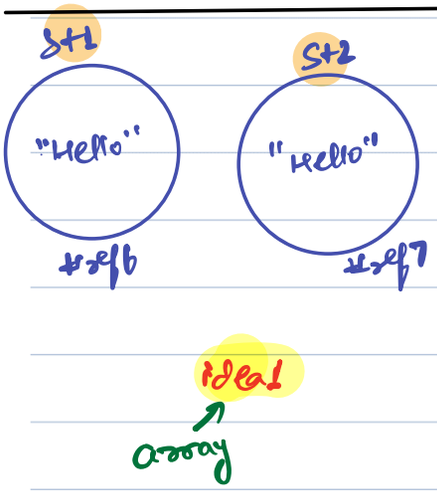


→ do not concatenate in String.

→ String St1 = "Hello";

String St2 = "Hello";





St1 = St1 + "e";

Save Space

↓

Point multiple same String to same address

↓

What if i change 1 String.

↓

Strings are immutable

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